

Artificial Intelligence — 10-Page Master Quick Revision

High-Yield Formula Sheet, Complexity Matrix & Top 10 BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Artificial Intelligence (CS24307)

Page 1: 38-Topic Master Syllabus Progress Checklist (Modules I – V)	Page 2: The Core AI Architecture: Agent Loop & 5 Architectural Types
Page 3: Uninformed Search: BFS, DFS, UCS, DLS & IDDFS Complexity Matrix	Page 4: Informed & Adversarial: A* Admissibility Proof, Minimax & Alpha-Beta
Page 5: Knowledge Representation: Propositional vs. First-Order Logic Translation	Page 6: Inference Engines: Forward Chaining, Backward Chaining & Resolution
Page 7: Automated Planning: STRIPS Action Schema & Goal Stack Mechanics	Page 8: Probabilistic AI: Bayes' Theorem & Bayesian Belief Networks Math
Page 9: Machine Learning: Neural Networks, Perceptron Activations & Backpropagation	Page 10: Generalization Theory: Underfitting vs. Overfitting & 9 Contrast Flashcards

The Complete 38–Topic Syllabus Master Checklist

38 / 38 Syllabus Topics Complete

Module	Topic #	Topic Name	Status
M1: Preliminaries (7)	1	What is AI? (4 Classical Approaches & Rational Agent Paradigm)	✔ Complete
	2	Evolution of AI (Turing 1950 to GenAI 2020s)	✔ Complete
	3	Intelligent Agents (Sense-Think-Act Loop & PEAS)	✔ Complete
	4	Concept of Rationality (Rationality vs. Omniscience)	✔ Complete
	5	Nature of Environments (O-D-E-D-C-N Taxonomy)	✔ Complete
	6	Structure of Agents (5 Agent Architectures)	✔ Complete
	7	Applications of AI (Healthcare, Robotics, Vision)	✔ Complete
M2: Search (6)	8	Search-Based Problem Solving (5-Tuple Formulation)	✔ Complete
	9	State-Space Search (BFS, DFS, UCS, DLS, IDDFS)	✔ Complete
	10	Tree Search vs. Graph Search (Explored Sets)	✔ Complete
	11	Heuristic Search (Greedy Best-First & A* Search)	✔ Complete
	12	Local Search (Hill Climbing & Simulated Annealing)	✔ Complete
	13	Complex Environments & Adversarial Game Trees (Minimax & Alpha-Beta)	✔ Complete
M3: Reasoning (9)	14	Knowledge-Based Agents (KB & Inference Engine)	✔ Complete
	15	Propositional Logic (Truth Tables & Equivalences)	✔ Complete
	16	Propositional to Predicate Logic Transition	✔ Complete
	17	Propositional Logic Agents (Wumpus World)	✔ Complete
	18	First-Order Predicate Logic (FOPL Quantifiers)	✔ Complete
	19	Knowledge Representation in FOL (English to Logic)	✔ Complete
	20	Forward Chaining (Data-Driven Modus Ponens)	✔ Complete
	21	Backward Chaining (Goal-Driven Hypothesis)	✔ Complete
	22	Resolution Refutation Inference Rule (CNF)	✔ Complete
M4: Planning (7)	23	Planning in AI (Goal-Directed Action Sequences)	✔ Complete
	24	Components of Planning Problems (STRIPS: PRE, ADD, DEL)	✔ Complete
	25	Types of Planning (State-Space, Partial-Order, HTN)	✔ Complete
	26	Goal Stack Planning (Subgoal Interleaving)	✔ Complete
	27	Reasoning Under Uncertainty (Axioms of Probability)	✔ Complete
	28	Bayesian Inference (Prior, Likelihood, Posterior)	✔ Complete
	29	Probabilistic Reasoning & Bayesian Belief Networks	✔ Complete
M5: Learning (9)	30	What is Learning? (Experience to Performance)	✔ Complete
	31	Rote Learning (Direct Memorization & Caching)	✔ Complete
	32	Learning by Taking Advice (Expert Rules)	✔ Complete

Module	Topic #	Topic Name	Status
	33	Learning from Examples (Inductive Supervised Learning)	✅ Complete
	34	Induction vs. Deduction in AI	✅ Complete
	35	Formal Learning Theory (Hypothesis Space, Bias-Variance)	✅ Complete
	36	Neural Net Learning (Perceptrons & Backpropagation)	✅ Complete
	37	Underfitting (High Bias, Low Capacity)	✅ Complete
	38	Overfitting (High Variance, Regularization & Dropout)	✅ Complete

🧠 Complete AI Master Algorithm Map

